

Forging
Neo-battalions.

SDTH
3-month

SDTH

3-month

Mission

Readying the Neo-Battalion

The next generation of ground operations will be fought by human-machine battalions: soldiers, sensors, robots, decoys, command layers, and edge intelligence operating as one system.

The challenge is to make that system trusted, coordinated, and useful when battlefield conditions break down.

Revamp the battalion stack from first principles - with edge AI, robotics, resilient coordination, and mission intelligence at its core.

Build a working system that helps ground forces command more effect in contested frontlines.

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A soldier in camouflage gear is operating a remote-controlled robotic vehicle (RCV) in a field. The RCV is a four-wheeled platform with a mounted machine gun. The soldier is holding a remote control device. The background is a grassy field.

Ukraine is already standing up robotic ground units for logistics, evacuation, mine tasks, and frontline support.

The battalion is becoming a synergistic human-machine system.

Today's autonomous innovations still break under:

1. degraded navigation
2. intermittent links
3. crowded spectrum
4. overloaded operators

**The next advantage is not one better robot.
It is a battalion stack that moves as one.**

We need a reliable backbone to support coordination among human, legacy, and non-legacy systems.

1 Beacon-Mesh PNT

City-scale operations need shared unit location and timing across soldiers, vehicles, sensors, and autonomous robots.

Build a deployable signal-beacon mesh that triangulates unit position and provides a shared timing layer across dense urban terrain.

2 Vision-only Localization

Urban terrain breaks traditional navigation sensors with canyons, occlusion, glass reflections, and multipath, especially for low-altitude drones and ground robots.

Build an onboard localization module that localizes primarily from live RGB feeds against urban structure in dense terrain.

We need command layers that let one operator control more systems, without losing trust or control.

3 UXV Orchestration

As UXVs scale across the battalion, the bottleneck shifts from platform availability to human command bandwidth.

Build a supervisory C2 layer that lets one operator task, redirect, and monitor multiple UAVs and UGVs.

4 UXV Trust Mesh

Future ground operations will mix friendly, hostile, unknown, and independently developed robotic systems in the same battlespace.

Build a low-signature trust layer for secure robotic identification and blue-on-blue prevention.

We need an edge-native stack that effectively incorporates soldiers into the command picture.

5 Battalion Edge S2

Forward units need local sensing and decision support, but cannot depend on server racks, cloud compute, or stable backhaul.

Build a low-SWaP edge S2 layer using distributed / soldier-carried compute and sensors.

6 Soldiers in C2

Autonomous command systems can track platforms, but often lack a live understanding of the soldiers still central to the fight.

Build a C2 layer that incorporates soldiers into autonomous command, starting with location, readiness, fatigue, injury risk, and taskability.

Build a mission layer of your choice: Identify 1–2 key state-of-the-art systems, and develop complete proofs of concept.

7 Composite Tactical Comms

Fiber gives reliability but not mobility; tactical wireless gives mobility but breaks under interference, terrain, and link loss.

Build a composite wireless backbone that combines relays, multi-path links, and degraded-mode protocols for fiber-like reliability.

8 Mission Refinement

Robotic missions produce valuable failure data that is lost after the mission. These same failures should be reproduced and reviewed in detail.

Build a mission-memory and 3D simulation layer that enables mission planning of the neo-battalion.

9 Battalion Counter-FPV

Fiber-optic FPV drones bypass RF jamming, creating a dominant, asymmetric frontline dynamic for both offense and defense.

Build an advanced system on a fiber-optic platform for battalion operations, exploring complex capabilities (e.g., multi-modal sensing, automated tasking).

Leverage sponsor tools



Agentic Software Development

Use Cognition's agentic software engineering workflows to accelerate code generation, debugging, test harnesses, simulation workflows, and software-heavy build components.

Auterion

Robotics & Drone Autonomy SDK

Build on Auterion's robotics autonomy stack to speed up mission planning, fleet control, vehicle integration, and operator-facing autonomy workflows.